

Economic and Social Impact of LAE

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Low Altitude Economics

Week 9

Learning Objectives

By the end of this class, you should be able to:

- **Build** a transparent market-size estimate for a defined Hong Kong LAE use case.
- **Distinguish** potential demand, serviceable demand, capacity, revenue, and value added.
- **Stress-test** a business case using scenarios, sensitivity analysis, and simple probability models.
- **Identify** who receives the benefits and who bears the costs of an LAE project.
- **Explain** how hedonic analysis and benefit-cost analysis may inform noise, access, and equity decisions.

From Commercial Value to Social Value

Imagine that several LAE services are operating across Hong Kong.

- Users may benefit from faster delivery, inspection, emergency response, or passenger travel.
- Nearby communities may experience noise, privacy concerns, visual effects, or unequal access to benefits.

A profitable service can still create social costs, while an unprofitable service may create important public benefits.

- **Quick poll (2 min):** Which matters most for public policy: operator profit, total social benefit, or how benefits and costs are distributed?

Session 1

Market Sizing, NASA Barriers & Uncertainty Modeling

Concept Primer: Market Sizing

Market sizing should connect the demand analysis from Week 4 with the capacity and cost analysis from later weeks.

- **Potential market:** All relevant customers or trips within a clearly defined geography, period, and use case.
- **Serviceable market:** The portion compatible with the proposed route, service level, willingness to pay, regulation, and infrastructure.
- **Obtainable demand:** The portion the operator may realistically serve after competition and capacity constraints.

TAM, SAM, and SOM are managerial labels, not a unique NASA methodology. NASA-sponsored UAM studies use detailed market models and examine multiple constraints.

Applied Tool: Defining the Potential Market

Start with the decision, not the largest available number.

1. Define the service, origin-destination market, customer, direction, and time period.
2. Estimate relevant existing trips or missions using transport, airport, logistics, public-service, or survey data.
3. Decide whether the project may divert existing activity, generate new activity, or change trip frequency.
4. Remove trips that cannot use the service because of geography, schedule, payload, accessibility, regulation, or other constraints.
5. Document data gaps and use ranges where evidence is weak.

Do not sum all trips to the airport unless they are genuinely relevant to the defined corridor and service.

Applied Tool: A Market Funnel with Capacity

Demand Funnel

$$SAM = TAM \times \text{Eligibility Share}$$

$$\text{Potential Operator Demand} = SAM \times \text{Expected Choice Share}$$

Capacity Check

$$\text{Served Demand} = \min\{\text{Potential Operator Demand}, \text{Reliable Capacity}\}$$

- Choice share should reflect price, generalized cost, service quality, and competition.
- Reliable capacity includes aircraft, seats or payload, infrastructure, operating hours, weather, and approvals.
- Avoid calling an unsupported capture percentage “realistic” without evidence.

Applied Tool: Revenue and Contribution

Revenue

$$\text{Daily Revenue} = \text{Served Units} \times \text{Average Revenue per Unit}$$

Contribution

$$\text{Daily Contribution} = \text{Daily Revenue} - \text{Variable Cost of Served Units}$$

- Use passenger-trips, deliveries, inspections, or missions consistently.
- Revenue is not cash flow or profit and does not by itself justify infrastructure investment.
- Compare contribution and longer-term cash flows with fixed operating cost and investment cost.

Back-of-the-Envelope: Filtering and Capacity

Illustrative airport-corridor assumptions:

Demand Funnel

- Relevant trips: 20,000/day
- Eligible time-sensitive segment: 10%
- $SAM = 20,000 \times 0.10 = 2,000$
- Expected choice share: 15%
- Potential operator demand: **300 passenger-trips/day**

Capacity Check

- 80 flights/day
- Four seats per flight
- Target load factor: 80%
- Reliable capacity:
 $80 \times 4 \times 0.80 = 256$
- **Served demand: 256 passenger-trips/day**

Takeaway: Obtainable demand is constrained by both customer choice and reliable operating capacity.

Illustrative assumptions; not current airport-corridor data.

Back-of-the-Envelope: From Served Demand to Contribution

Assumptions: 256 passenger-trips/day, average fare HK\$400, variable cost HK\$120/passenger-trip, and 330 operating days/year.

Daily Result

- Revenue:
 $256 \times 400 = \text{HK\$}102,400$
- Variable cost:
 $256 \times 120 = \text{HK\$}30,720$
- **Daily contribution: HK\$71,680**

Annual Result

- Annual revenue: HK\$33.8M
- Annual variable cost: HK\$10.1M
- **Annual contribution: about HK\$23.7M**
- Fixed cost and investment remain to be covered.

Illustrative assumptions; excludes fixed cost, financing, tax, and demand variation.

Concept Primer: Constraints Identified in UAM Market Studies

The NASA-sponsored UAM market study examined interacting constraints rather than a universal list of five revenue haircuts.

- legal and regulatory requirements;
- certification and safety;
- infrastructure and airspace capacity;
- weather and technology limitations;
- public perception and community acceptance; and
- high near-term service cost and other economic constraints.

Managerial lesson: Each constraint may affect demand, capacity, price, cost, timing, or probability of launch through a different channel.

Applied Tool: Constraint Mapping

Constraint	Primary model channel
Regulatory delay	Launch date, cost, or probability
Infrastructure scarcity	Reliable capacity and access
Public acceptance	Choice share, hours, or route
Weather	Serviceable capacity and reliability
Technology maturity	Cost, payload, range, and downtime
Competition	Price and operator choice share

Avoid multiplying arbitrary independent “haircuts” when the same effect is already reflected elsewhere. For example, weather should not reduce both operating days and revenue again unless the channels are distinct.

Back-of-the-Envelope: Explicit Constraint Channels

Base assumptions: 300 potential passenger-trips/day, fare HK\$400, and 365 calendar days.

Demand and Capacity

- Reliable capacity: 256 trips/day
- Acceptance scenario: 90% of potential demand
- Adjusted demand: 270 trips/day
- Served trips: $\min\{270, 256\} = 256$

Operating Days and Revenue

- Serviceable days: 330/year
- Annual revenue: $256 \times 400 \times 330$
- **HK\$33.8M**
- Technology and compliance costs enter the cost model rather than another revenue haircut.

Takeaway: Model each constraint through its actual channel and check for double counting.

Illustrative assumptions for classroom analysis.

Session 2

Uncertainty, Monte Carlo Sensitivity & Macro Impact

Concept Primer: Uncertainty

Uncertainty means that important inputs and outcomes cannot be known precisely when a decision is made.

- demand and willingness to pay;
- approval timing and operating conditions;
- weather and serviceable capacity;
- technology performance and maintenance;
- public acceptance and competition; and
- prices, costs, and financing conditions.

Managerial lesson: Use a base case, downside and upside scenarios, sensitivity analysis, and probability ranges where justified. Not every uncertainty is measurable or hedgeable.

Concept Primer: Monte Carlo Simulation

A Monte Carlo simulation repeatedly samples uncertain inputs from specified probability distributions and recalculates the model.

- It produces a distribution of possible outcomes rather than one forecast.
- Results depend on the chosen distributions, correlations, model structure, and data quality.
- Correlated inputs matter. Weather may affect capacity, demand, price, and cost simultaneously.
- Useful outputs include median NPV, downside percentiles, probability of a cash shortfall, and major risk drivers.

Simulation does not make weak assumptions accurate; it makes their implications visible.

Applied Tool: Expected Value and Downside Risk

Expected Profit

$$E[\Pi] = \sum_s \text{Pr}(s) \Pi_s$$

- Expected value is the probability-weighted average across scenarios.
- It does not reveal how severe the downside can be or whether the business has enough liquidity to survive it.
- Also examine the probability of loss, worst credible cash shortfall, and a downside percentile such as the 10th percentile.
- Required cash reserves depend on the timing of cash flows, financing access, risk tolerance, and recovery plan, not expected value alone.

Back-of-the-Envelope: Expected Profit and Probability of Loss

Assumption: Annual cost = HK\$5M.

Three Scenarios

- Strong: 20%, revenue HK\$10M, profit HK\$5M
- Base: 50%, revenue HK\$8M, profit HK\$3M
- Weak: 30%, revenue HK\$4M, loss HK\$1M

Risk Summary

- Expected revenue: HK\$7.2M
- **Expected profit: HK\$2.2M**
- **Probability of loss: 30%**
- Minimum cash needed in the weak scenario depends on when costs and revenue occur.

Takeaway: A positive expected profit can coexist with a material probability of loss.

Illustrative scenario probabilities and values.

A HK\$350 Billion Projection and Distributional Questions

A 2025 report by Our Hong Kong Foundation projects that Hong Kong's LAE-related industries could generate approximately HK\$350 billion in annual industry value added by 2050, with supporting services contributing the majority.

- This is a scenario-based projection, not government revenue, operator revenue, GDP automatically created, or a guaranteed outcome.
- Value added measures the contribution of labour and capital after subtracting intermediate inputs; it should not be added directly to market revenue estimates.
- Possible benefits include productivity, logistics, public-service performance, employment, research, finance, insurance, legal, and professional services.
- Distribution matters: who gains access, who receives employment and business opportunities, and who bears noise, privacy, safety, land-use, or fiscal costs?

Session 3

Stakeholder Effects, Hedonic Valuation & Equity

Concept Primer: Hedonic Valuation

Hedonic analysis estimates how observable attributes are associated with market prices while controlling for other factors.

- For housing, attributes may include size, age, location, transport access, schools, amenities, and environmental conditions such as noise.
- A credible study needs many comparable transactions and a strategy for separating noise effects from other site characteristics.
- A simple difference between two homes does not establish the causal value of noise.
- Property-price effects capture capitalization in housing markets, not every health, annoyance, productivity, or distributional effect.

Applied Tool: A Hedonic Price Model

Illustrative Specification

$$\ln(P_i) = \alpha + \beta N_i + \gamma A_i + \delta X_i + \varepsilon_i$$

- P_i = property price or rent.
- N_i = measured noise exposure.
- A_i = access to the relevant service or transport network.
- X_i = other property and neighbourhood characteristics.
- β estimates the conditional association between noise and price; causal interpretation requires stronger research design.

Managerial use: Inform benefit-cost analysis and mitigation, not decide community acceptance by property value alone.

Back-of-the-Envelope: Illustrative Property-Value Scenarios

Assumption: Baseline apartment value = HK\$8M. The percentages below are hypothetical, not estimates for drone noise in Hong Kong.

Higher Noise, Limited Local Access

- Assumed noise effect: -4%
- Assumed access effect: 0%
- New value: $8M \times 0.96$
- **HK\$7.68M**

Lower Noise, Useful Local Access

- Assumed noise effect: -1%
- Assumed access effect: $+3\%$
- New value: $8M \times 1.02$
- **HK\$8.16M**

Takeaway: The arithmetic illustrates competing effects. Actual coefficients require local transaction, noise, access, and neighbourhood data.

Social Equity and Distributional Analysis

A social-impact assessment should ask who receives each benefit and who bears each cost.

- **Access:** affordability, geographic coverage, accessibility, digital access, and service availability.
- **Exposure:** noise, privacy, visual impact, safety risk, construction, and land-use effects.
- **Opportunity:** jobs, training, procurement, entrepreneurship, and supporting-service development.
- **Public finance:** subsidies, infrastructure spending, availability payments, and alternative uses of public funds.
- **Public services:** emergency response and medical logistics may create broad benefits, but should be evaluated against alternative delivery methods.

ESG reporting can support transparency, but a project should define measurable outcomes rather than assume that an EMS label guarantees equity.

Comparing Social Impact Across Places

Social impacts depend on local context and should not be ranked through broad claims.

- Hong Kong's dense urban environment makes access, noise exposure, land use, existing public transport, and distributional effects especially important.
- Different GBA cities may have different trip patterns, industrial structures, land availability, public-service needs, and regulatory settings.
- International evidence on aviation noise and property values can inform hypotheses, but coefficients should not be transferred mechanically to drone or eVTOL operations in Hong Kong.
- A common evaluation framework can compare safety, time savings, emissions, noise, accessibility, employment, fiscal cost, and distribution across stakeholder groups.

In-Class Exercise: Market, Risk, and Social Impact

Activity: Stress-Testing the Business and Public Case

- **Format:** Groups of 4–5 with the spreadsheet template
- **Tasks:**
 1. define relevant trips, eligibility, expected choice share, and reliable capacity;
 2. calculate served demand, revenue, variable cost, and contribution;
 3. vary at least three uncertain inputs and report expected profit and probability of loss;
 4. identify one benefit and one cost for each major stakeholder group; and
 5. test one mitigation and state who pays for it.
- **Timing:** 15 minutes plus 5-minute discussion.

Exercise Template: Integrated Dashboard

Section	Inputs and outputs
Market	Relevant trips, eligibility, choice share
Capacity	Flights, capacity per flight, serviceable days
Economics	Price, variable cost, fixed cost, contribution
Uncertainty	Ranges, distributions, correlations, probability of loss
Social impact	Benefits, costs, affected groups, mitigation

Do not use unexplained barrier toggles. Each scenario assumption should change a specific model input and include a source or justification.

Key Takeaways

- Market size must be defined by service, corridor, customer, period, choice behaviour, and reliable capacity.
- Revenue, contribution, profit, value added, and social benefit are different concepts.
- Constraints should enter through explicit demand, capacity, price, cost, timing, or probability channels without double counting.
- Expected value is useful, but decision-makers also need probability of loss, downside cash needs, and key risk drivers.
- Hedonic analysis requires comparable data and careful identification; simple assumed property discounts are scenarios, not evidence.
- Social evaluation must consider total benefits and costs together with their distribution across users, communities, workers, taxpayers, and businesses.

Week 9 Assignment Briefing

Market, Risk, and Social-Impact Memo with Model (maximum 1,200 words plus spreadsheet)

- **Due:** Beginning of Week 10 (group assignment)
- **Requirements:**
 1. define the relevant market using sourced data and transparent exclusions;
 2. estimate serviceable and served demand after a reliable-capacity check;
 3. calculate revenue, variable cost, contribution, and relevant fixed cost;
 4. model at least three uncertain inputs, including correlations where material;
 5. report expected profit, probability of loss, and one downside cash-flow measure;
 6. identify benefits, costs, and distributional effects for at least four stakeholder groups; and
 7. evaluate one mitigation, including its cost, beneficiary, and payer.
- **Grading focus:** Model consistency (30%), evidence and assumptions (25%), risk analysis (20%), social-impact analysis (15%), professionalism (10%).

Sustainability Challenges in Urban Air Operations

Preview question for next week:

We just quantified the impact of noise and emissions... but what is the actual mathematical cost of solving them?

Next Week Preview:

Green Skies: Carbon, Noise, and the Cost of Clean Tech

Pre-Class Readings Reminder (for Week 10):

- *NASA UAM Market Study (Sustainability metrics)*
- *Our Hong Kong Foundation (ESG mandates for LAE)*

Questions?

(Final 15 minutes reserved for extended Q&A)